Advancing Wildfire Robotics with the FIReVision Simulator

Wildfires are increasing in frequency and complexity, necessitating advanced technological solutions.





FIReVision,
a cutting-edge photorealistic thermal infrared simulator.

Uses AirSim and Unreal Engine 5 to help improve the capabilities of unmanned aerial systems (UAS) in wildfire monitoring and management.



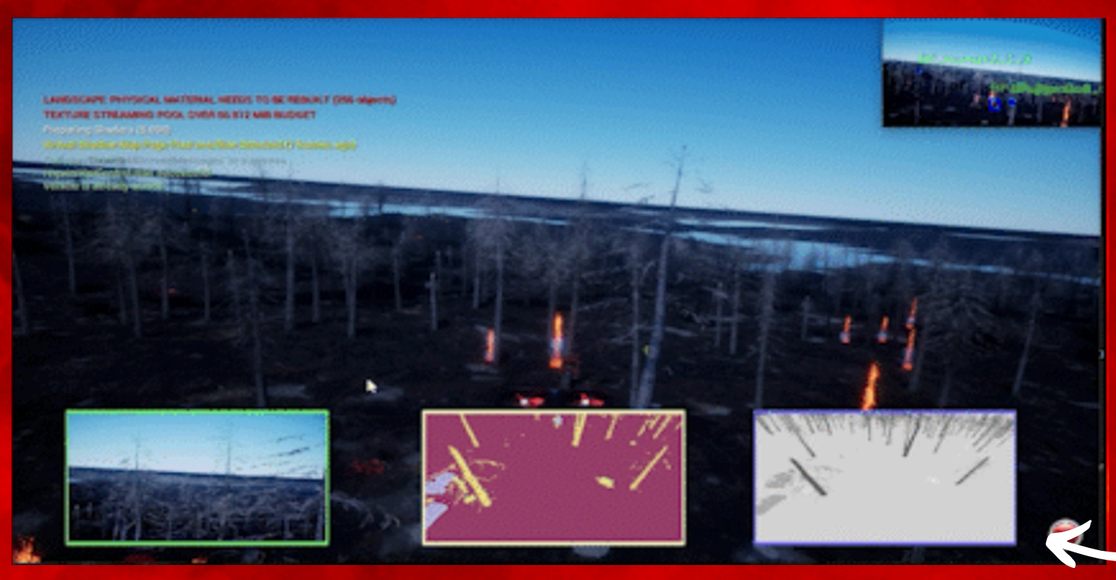
Existing UAS Wildfire Simulators

- Limited in capabilities and realism.
- Restricted to 2D space.
- Lack realistic UAS dynamics.
- Do not include important targets like crew and vehicles
- Usually lack a camera-based perception system.

FIReVision

- Realistic RGB and Thermal Vision.
- Visually realistic, dynamic smoke.
- Poses challenges by being visually degrading, hazardous, and dynamic.
- Automatic Data Labeling with Object Detection & Depth Estimation.
- Is Open-Source & Accessible

Personal Contributions



I developed Python scripts for:

- **Drone Navigation:** Keyboard control system for simulated drone operation.
- **Real-Time Object Detection:** Enhances situational awareness within simulated environments.
- **Data Collection:** Facilitates the generation of diverse image datasets crucial for object detection research.